

```
In [1]: #Python input()
```

```
In [2]: # 1. What is the input() function in Python used for?
```

```
In [3]: # Read from user, default is string.
```

```
In [4]: # 2 . How can you accept an integer as input from the user using input()?
```

```
In [5]: int(input("Enter a string: "))
```

```
Out[5]: 304
```

```
In [ ]: # 3. How do you accept a float input from the user?
```

```
In [6]: float(input("Enter a string: "))
```

```
Out[6]: 98.7
```

```
In [ ]: # 4. How can you take multiple space-separated values as input?
```

```
In [7]: input("Enter a string: ").split()
```

```
Out[7]: ['My', 'name', 'is', 'Rashmi']
```

```
In [ ]: # 5. How do you check if a number entered by the user is positive, negative, or zero?
```

```
In [13]: num = int(input("Enter the number :"))

if num >0:
    print("Positive")
elif num<0:
    print("Negative")
else:
    print("Zero")
```

Negative

```
In [ ]: # 6. How do you convert user input to a list of integers?
```

```
In [17]: input("Enter integers: ").split()
```

```
Out[17]: ['98', '76', '678']
```

```
In [ ]: # 7. How do you accept a string input and print it in uppercase?
```

```
In [19]: input("Enter string :").upper()
```

```
Out[19]: 'SAANCHI'
```

```
In [ ]: # 8. Write a Python program that accepts a string and prints the number of vowels in it.
```

```
In [20]: text = input("Enter string :")
vowels = 'aeiou'
count = sum(1 for char in text if char.lower() in vowels)
print("Vowels", count)
```

Vowels 2

```
In [ ]: # 9. Write a program that takes a number as input and checks if it is even or odd.
```

```
In [22]: num = int(input("Enter number: "))
if (num % 2 == 0):
    print("Even number")
else:
    print("Odd number")
```

Odd number

```
In [ ]: # 10. How would you check if a string is a palindrome using `input()`?
```

```
In [28]: text = input("Enter string :")
palin_text = text[::-1]
if (text == palin_text):
    print("Palindrome number")
else:
    print("Not a palindrome")
```

Palindrome number

```
In [ ]: # 11. Write a program that takes a number as input and prints its square.
```

```
In [31]: num = int(input("Enter number : "))  
print("Square of number :", num**2)
```

Square of number : 16

```
In [ ]: # 12. Write a program that asks for a number and prints whether it is divisible by 3.
```

```
In [33]: num = int(input("Enter number : "))  
div_num = num %3  
if (div_num == 0):  
    print("Divisible by 3")  
else:  
    print("Not divisible by 3")
```

Not divisible by 3

```
In [ ]: # 13. How would you check if a number is divisible by both 3 and 7?
```

```
In [36]: num = int(input("Enter number : "))  
div_num1 = num %3  
div_num2 = num %7  
if (div_num1 == 0 and div_num2 ==0):  
    print("Divisible by 3 & 7")  
else:  
    print("Not divisible by 3 & 7")
```

Divisible by 3 & 7

```
In [ ]: # 14. How do you accept a list of comma-separated values as input?
```

```
In [42]: num = input("Enter numbers :").split(',')  
num
```

```
Out[42]: ['Rash', 'mi', 'san', 'nchi']
```

```
In [ ]: #15. Write a Python program that takes two numbers as input and prints their product.
```

```
In [43]: num1 = int(input("Enter number 1 : "))
num2 = int(input("Enter number 2 : "))
product = num1 * num2
print(product)
```

200

```
In [ ]: # 16. Write a program that checks if the input number is a prime number.
```

```
In [45]: num = int(input("Enter number : "))
if num > 2:
    for i in range(2,num):
        if num % i == 0:
            print("Not a Prime")
            break
        else:
            print("Prime")
else:
    print("Not a prime")
```

Not a Prime

```
In [ ]: # 17. How can you accept a boolean value (True/False) from the user?
```

```
In [47]: input("Enter True or False :").lower()
```

```
Out[47]: 'false'
```

```
In [ ]: # 18. Write a program that accepts a string and prints the reverse of that string.
```

```
In [48]: print(input("Enter string :")[::-1])
```

DCBA

```
In [ ]: # 19. Write a program that asks for a user's name and age and prints a message.
```

```
In [52]: name = input("Enter name : ")
age = int(input("Enter age : "))
print("Hello ",name,"!\nYour age is ",age)
```

Hello Rashmi !
Your age is 35

In []: *# 20. Write a program to calculate the factorial of a number using `input()`.*

```
In [55]: num = int(input("Enter a number: "))
factorial = 1
for i in range(1, num + 1):
    factorial *= i
    i = i+1
print("Factorial:", factorial)
```

Factorial: 24

In []: *# 21. How do you prevent a user from entering an empty string?*

```
In [62]: num = input("Enter a string: ")
empty_num = num.strip()
if not empty_num:
    print("Empty")
else:
    print("Not empty")
```

Empty

In []: *# 22. Write a program to check if an entered number is a perfect square.*

```
In [74]: import numpy as np
num = int(input("Enter a number: "))
sqrt_num = np.sqrt(num)
square_perfect = sqrt_num **2
if (num == square_perfect):
    print("Perfect Square root")
else:
    print("Not a square root")
```

Perfect Square root

In []: *# 23. Write a program that asks the user for a year and determines if it's a Leap year.*

```
In [76]: year = int(input("Enter a year: "))
if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
```

```
    print("Leap year")
else:
    print("Not a leap year")
```

Leap year

```
In [ ]: # 24. How can you remove leading and trailing spaces from a string input?
```

```
In [78]: input("Enter a string: ").strip()
```

```
Out[78]: 'Rashmi'
```

```
In [ ]: # 25. How do you handle incorrect inputs when you expect an integer using `input()`?
```

```
In [79]: try:
          num = int(input("Enter a number: "))
        except ValueError:
          print("Invalid input! Please enter a valid integer.")
```

Invalid input! Please enter a valid integer.

```
In [ ]: # 26. Write a program that accepts a string and counts the occurrence of a particular character.
```

```
In [81]: text = input("Enter a string: ")
          char = input("Enter a character to count: ")
          text.count(char)
```

```
Out[81]: 4
```

```
In [ ]: # 27. How would you convert user input to lowercase using `input()`?
```

```
In [82]: input("Enter a string: ").lower()
```

```
Out[82]: 'saanchibalurkar'
```

```
In [ ]: # 28. Write a program that accepts a number and prints whether it is a multiple of 10.
```

```
In [83]: num = int(input("Enter a number: "))
          if num % 10 == 0:
            print("Multiple of 10")
```

```
else:
    print("Not multiple of 10")
```

Multiple of 10

```
In [ ]: # 29. How would you check if a string contains only alphabets using `input()`?
```

```
In [86]: text = input("Enter a string: ")
if text.isalpha():
    print("Only alphabets")
else:
    print("Not all alphabets")
```

Not all alphabets

```
In [ ]: # 30. Write a program to count the number of words in a sentence entered by the user.
```

```
In [87]: text = input("Enter a sentence: ")
len(text.split())
```

Out[87]: 5

```
In [ ]: # 31. How would you accept a date input from the user in Python?
```

```
In [90]: from datetime import datetime
date_str = input("Enter a date (YYYY-MM-DD): ")
date = datetime.strptime(date_str, "%Y-%m-%d")
print("Entered date:", date)
```

Entered date: 2025-09-11 00:00:00

```
In [ ]: # 32. Write a program that checks if the entered number is divisible by both 3 and 5.
```

```
In [94]: num = int(input("Enter number : "))
div_num1 = num %3
div_num2 = num %5
if (div_num1 == 0 and div_num2 ==0):
    print("Divisible by 3 & 5")
else:
    print("Not divisible by 3 & 5")
```

Divisible by 3 & 5

```
In [ ]: # 33. Write a program to swap the values of two variables using `input()`.
```

```
In [95]: a = input("Enter first value: ")
b = input("Enter second value: ")
a, b = b, a
a,b
```

```
Out[95]: ('20', '10')
```

```
In [ ]: # 34. Write a program to take user input and print it without spaces between words.
```

```
In [96]: text = input("Enter a string: ")
print(text.replace(" ", ""))
```

MyNameisRashmiBalurkar

```
In [ ]: # 35. How do you validate if an entered input is a valid email address?
```

```
In [99]: import re                # regular expressions module
email = input("Enter email: ")
if re.match(r"^[^@]+@[^@]+\.[^@]+$", email):
    print("Valid email")
else:
    print("Invalid email")
```

Valid email

```
In [ ]: # 36. Write a program that accepts a number and prints its cube.
```

```
In [100... num = int(input("Enter a number: "))
num**3
```

```
Out[100... 27
```

```
In [ ]: # 37. How would you accept and store multiple names from the user?
```

```
In [106... names = input("Enter multiple names: ").split(',')
names
```

```
Out[106... ['Rashmi', 'Sidhesh', 'Saanchi']
```

```
In [ ]: # 38. How would you extract numbers from a string entered by the user?
```

```
In [108... import re
text = input("Enter a string: ")
numbers = re.findall(r'\d+', text)
print("Extracted numbers:", numbers)
```

Extracted numbers: ['075', '98']

""import re → loads the regular expression module. r'\d+' → means: \d → any digit (0–9)

- → one or more occurrences of digits re.findall() → returns a list of all matches in the input string.

```
In [ ]: # 39. How do you find the maximum number from a list of integers entered by the user?
```

```
In [117... numbers = list(map(int, input("Enter numbers separated by spaces: ").split()))
print("Maximum number:", max(numbers))
```

Maximum number: 765

```
In [ ]: # 40. How would you prompt the user for input until they enter a valid number?
```

```
In [119... while True:
    try:
        num = int(input("Enter a valid number: "))
        break
    except ValueError:
        print("Invalid input, please enter a number.")
```

Invalid input, please enter a number.

Invalid input, please enter a number.

```
In [ ]: # 41. Write a program to check if the entered string has digits.
```

```
In [120... text = input("Enter a string: ")
if any(char.isdigit() for char in text):
    print("Contains digits")
else:
    print("No digits")
```

Contains digits

```
In [ ]: # 42. Write a program to check if the entered string has only whitespace characters.
```

```
In [123... text = input("Enter a string: ")
if text.isspace():
    print("Only whitespace")
else:
    print("Contains non-whitespace characters")
```

Only whitespace

```
In [ ]: # 43. Write a program to find the sum of all digits in a string entered by the user.
```

```
In [124... text = input("Enter a string: ")
digit_sum = sum(int(digit) for digit in text if digit.isdigit())
print("Sum of digits:", digit_sum)
```

Sum of digits: 26

```
In [ ]: # 44. Write a program that accepts a number and prints its absolute value.
```

```
In [126... abs(int(input("Enter number: ")))
```

Out[126... 76

```
In [ ]: # 45. How would you check if a string entered by the user contains any uppercase Letters?
```

```
In [128... text = input("Enter a string: ")
if any(char.isupper() for char in text):
    print("Contains uppercase")
else:
    print("No uppercase")
```

Contains uppercase

In []: *# 46. Write a program that converts Celsius to Fahrenheit.*

```
In [130... celsius = int(input("Enter temperature in celsius: "))
fahrenheit = (celsius * 9/5) + 32
print(f"Temperature in fahrenheit: {fahrenheit}")
```

Temperature in fahrenheit: 69.8

In []: *# 47. Write a program to find the average of a list of numbers entered by the user.*

```
In [132... numbers = list(map(int, input("Enter numbers separated by space: ").split()))
print("Average:", sum(numbers) / len(numbers))
```

Average: 33.333333333333336

In []: *# 48. Write a program to count the number of consonants in a string entered by the user.*

```
In [133... text = input("Enter a string: ")
consonants = "bcdfghjklmnpqrstvwxyz"
count = sum(1 for char in text.lower() if char in consonants)
print("Number of consonants:", count)
```

Number of consonants: 4

In []: *# 49. How do you check if a string entered by the user contains any punctuation?*

```
In [136... import string
text = input("Enter a string: ")
if any(char in string.punctuation for char in text):
    print("Contains punctuation")
else:
    print("No punctuation")
```

Contains punctuation

In []: *# 50. Write a program that accepts a sentence and prints the Longest word.*

```
In [139... text = input("Enter a sentence: ")
words = text.split()
longest_word = max(words, key=len)
print("Longest word:", longest_word)
```

Longest word: Rashmi

In []: